Celestial Mechanics: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

15 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Celestial Mechanics: worked solutions

Solution 1 — A different planet. Return →

Idea. Compare ratios so Earth's mass and radius cancel. $g'/g = (M'/M)/(R'/R)^2 = 2/9$, so $g' = 0.222g$. Escape speed scales as $\sqrt{M/R}$, giving $v'_{\rm esc}/v_{\rm esc} = \sqrt{2/3} = 0.816$. For Earth's 11.2 km/s this is **9.13 km/s**. A larger planet can have weaker surface gravity if its mass does not grow fast enough.

Check. Both results are below Earth's; the radius increase wins in both scalings.

Solution 2 — One Earth radius up. Return →

The orbital radius is $r = 2R_\oplus$, not $R_\oplus$. Thus $g' = g_0/4 = \mathbf{2.45\ m\,s^{-2}}$. Using $GM = g_0 R_\oplus^2$,

$$v = \sqrt{g_0 R_\oplus/2} = 5.59\ \mathrm{km\,s^{-1}}, \qquad T = 2\pi r/v = 1.436\times10^4\ \mathrm{s} = 3.99\ \mathrm{h}. \tag{1}$$

Check. A higher orbit is slower and has a longer period than a near-surface orbit.

Solution 3 — A comet's two speeds. Return →

In AU–yr–solar-mass units, $T = \sqrt{4^3} = \mathbf{8\ yr}$. The distances are $r_p = 4(0.25) = \mathbf{1\ AU}$ and $r_a = 4(1.75) = \mathbf{7\ AU}$. With $v_0 = 29.7847$ km/s,

$$v_p = v_0\sqrt{\frac{1+e}{a(1-e)}} = 39.40\ \mathrm{km\,s^{-1}}, \quad v_a = v_0\sqrt{\frac{1-e}{a(1+e)}} = 5.629\ \mathrm{km\,s^{-1}}. \tag{2}$$

Check. $r_p v_p = r_a v_a$; the speed ratio is 7, the inverse of the distance ratio.

Solution 4 — The hidden wobble. Return →

The period uses total mass: $T = \sqrt{3^3/3} = \mathbf{3\ yr}$. The barycentre divides the separation inversely with mass: the $2M_\odot$ star has $r_1 = 1$ AU and the $1M_\odot$ star $r_2 = 2$ AU. Their speeds are $2\pi r/T$, namely 2.094 and 4.189 AU/yr. Since 1 AU/yr = 4.74047 km/s, these are **9.93 and 19.9 km/s**.

Check. $m_1 v_1 = m_2 v_2$ in the barycentric frame.

Solution 5 — Weighing an eclipsing binary. Return →

Idea. The semiamplitude ratio gives the mass ratio; the sum gives the physical scale. $m_2/m_1 = K_1/K_2 = 0.5$. For an edge-on circular binary, $a = (K_1 + K_2)P/(2\pi)$, so

$$m_1 + m_2 = \frac{P(K_1+K_2)^3}{2\pi G} = 6.04 M_\odot. \tag{3}$$

Therefore $m_1 = 4.03 M_\odot$ **and** $m_2 = 2.01 M_\odot$. The star with the smaller velocity is the heavier one.

Check. The ratio is 2:1 and the sum is 6.04. No inclination correction is needed because $i = 90°$.

Solution 6 — At the semimajor axis. Return →

Substitute $r = a$: $v^2 = GM(2/a - 1/a) = GM/a$. The result is independent of e because the total energy depends only on a. Eccentricity fixes angular momentum and therefore the velocity's direction and the orbit's shape. Equal speed at one radius does not imply equal velocity or a circular path.

Check. For $e = 0$, the body stays at $r = a$; for $e > 0$, it crosses that radius while moving inward or outward.

Solution 7 — An ideal Earth–Mars transfer. Return →

Idea. Use the transfer ellipse at both endpoints. $a_t = (1 + 1.524)/2 = \mathbf{1.262\ AU}$ and $t = \frac{1}{2}\sqrt{a_t^3} = \mathbf{258.9\ days}$. With $v_0 = 29.7847$ km/s,

$$\Delta v_1 = v_0\left(\sqrt{\frac{2(1.524)}{2.524}} - 1\right) = 2.95\ \mathrm{km\,s^{-1}}, \tag{4}$$

$$\Delta v_2 = \frac{v_0}{\sqrt{1.524}}\left(1 - \sqrt{\frac{2}{2.524}}\right) = 2.65\ \mathrm{km\,s^{-1}}. \tag{5}$$

The ideal total is **5.60 km/s**. **Check.** Both burns speed the craft up heliocentrically. This is not the launch cost from Earth's surface.

Solution 8 — A sideways boost. Return →

Let $v_c^2 = GM/r_0$. Then $\epsilon = (1.44 - 2)v_c^2/2 = \mathbf{-0.28GM/r_0}$. Since $\epsilon = -GM/(2a)$, $a = r_0/0.56 = \mathbf{1.786r_0}$. The burn point is periapsis, so $e = 1 - r_0/a = \mathbf{0.44}$ and $r_a = a(1 + e) = \mathbf{2.571r_0}$.

Check. The speed is below $\sqrt{2}v_c$, so the orbit remains bound. A tangential burn at an apsis leaves that point an apsis.

Solution 9 — Can we separate the stars? Return →

By the definition of a parsec, 1 AU at 1 pc subtends 1 arcsecond. Thus $\theta = 2/100 = \mathbf{0.020''}$, or 20 mas, for a favourably projected separation. The period is $\sqrt{2^3/2} = \mathbf{2\ yr}$.

Check. Projection can reduce the apparent separation; the distance is essential. Angular separation alone would not give a mass.

Solution 10 — A moon too close. Return →

$d_R = 2.44R_M(1.5/3.0)^{1/3} = 2.44(0.7937)R_M = \mathbf{1.94R_M}$. The rigid strengthless coefficient 1.26 would give exactly $1.00R_M$ for this density ratio. A fluid body deforms more readily, so the ideal fluid disruption scale is farther out.

Check. These are model estimates. Real cohesion can allow a small solid body to survive inside the fluid limit.

Solution 11 — The Earth–Sun neighbourhood. Return →

The Hill estimate is $a(q/3)^{1/3} = 1\,\mathrm{AU} \times (10^{-6})^{1/3} = \mathbf{0.0100\ AU} = \mathbf{1.50 \times 10^{6}\ km}$. This is an approximation, not the exact L1 distance. Since $3 \times 10^{-6} < 0.03852$, the ideal circular restricted three-body problem satisfies the L4/L5 linear-stability criterion.

Check. L1 is not a zero-gravity point; it co-rotates with Earth.

Solution 12 — Density from a low orbit. Return →

Set $r = R$ and $M = 4\pi R^3 \rho/3$ in $T^2 = 4\pi^2 R^3/(GM)$. The radius cancels, giving $\rho = 3\pi/(GT^2)$. With $T = 5400$ s, $\rho = 4.84 \times 10^{3}\ \mathrm{kg\,m^{-3}}$.

Check. The result has density units. A real orbit above the surface would make this simple surface-radius estimate an approximation.

Solution 13 — Moon versus Sun in the tides. Return →

Idea. Compare M/d^3, not M/d^2. Using the centre-to-near-side convention $2GMR/d^3$ gives $\Delta g_M = \mathbf{1.10 \times 10^{-6}\ m\,s^{-2}}$ and $\Delta g_\odot = \mathbf{5.05 \times 10^{-7}\ m\,s^{-2}}$. Their ratio is **2.18**. The Sun's direct gravitational pull on Earth is much larger, but its greater distance makes its field more nearly uniform across Earth.

Check. Tides depend on a gradient. Actual ocean tides also depend on basin geometry and resonance.

Solution 14 — Borrowing a planet's motion. Return →

In the planet frame the initial relative velocity is -10 km/s and the final one $+10$ km/s. Add the planet's 13 km/s to return to the heliocentric frame: initial 3 km/s, final **23 km/s**, a gain of **20 km/s**. The ideal 180° turn gives the maximum change $2V_p$.

Check. Relative speed is unchanged in this ideal planet-frame encounter; heliocentric kinetic energy is not. The planet loses an immeasurably small amount of orbital energy and momentum. A real turn is less than 180°.

Solution 15 — The inclination degeneracy. Return →

Idea. The observables constrain a combination of companion mass and inclination. Substitution gives $f = \mathbf{0.12475 M_\odot}$. Let $x = m_2/M_\odot$ and $m_1 = 1 M_\odot$. For $i = 90^\circ$, solve $x^3/(1+x)^2 = 0.12475$, yielding $x = 0.716$. For $i = 30^\circ$, the right-hand side is $f/\sin^3 i = 0.99797$, yielding $x = 2.145$.

Check. A smaller inclination requires a larger companion to produce the same observed wobble. The minimum is not a measurement of the actual mass unless the inclination is known. This is the standard spectroscopic mass-function degeneracy.